Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

LastMile Response

Team name: MOB

Team leader name: Lithira Aponsu & Pulsara Aponsu

Problem Statement: Sustainability & Social Impact

Brief about the idea

When the rain doesn't stop, every minute counts.



A family waits at a temple with no food. A shelter runs out of water. The help exists, but it is lost in phone calls, letters and messages between a dozen offices, and the people affected hear last.

LastMile Response puts every agency, every request and every person on one live portal, with Gemini doing the paperwork.

One live picture
for every agency, the moment it changes

Say what's needed
the AI prepares it; a person approves

Anyone can ask
in Sinhala, Tamil or English, and see help coming

Prototype on Google Cloud with Gemini. Fictional exercise data; no live dispatch, message delivery or payments.

Opportunities

A cry for help deserves an answer, not a phone tree

01

What makes it different?

One live record, not twenty phone calls

Public requests, requests between agencies, warnings, maps, shelters, stock and the resident's tracker are one connected record. AI works inside the record, not as a chatbot on the side.

02

How does it solve the problem?

From “we need water” to “delivered”

Each need goes to the body that owns it. Every step notifies the other side, overdue requests escalate, and the resident sees progress with a tracking code.

03

What is the USP?

AI that prepares. People who decide.

A Google ADK agent with ten allow-listed tools turns one sentence into an exact plan. Code checks numbers and permissions, a person approves the exact version, and every step is recorded.

Sri Lanka focus: 25 districts and their DS divisions, in English, Sinhala and Tamil.

List of features offered by the solution

Everything a response needs, in one place

Ask for help, any language

AI HELPS

Words, photos and area from the public page, no login. A tracking code shows each step.

Public requests board

AI HELPS

Gemini reads need, people and urgency; code checks it; spam is set aside only when two readers agree.

Say it in chat

AI HELPS

One sentence becomes an exact plan: a request, a letter, a status update. Nothing applies until approved.

Requests between agencies

Accept, deliver, confirm; priorities, due times and automatic escalation up to the DMC.

Warnings and live map

Read figures and NBRO-style warnings from bulletins; the map colours every DS division.

Texts in three languages

AI HELPS

Warning texts in English, Sinhala and Tamil with the facts locked; a person approves.

Notifications bell

What is waiting for your role, since when, and what other bodies did; read each or all.

Shelters, stock and who-does-what

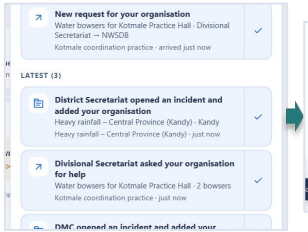
Shared stock holds, shelter capacity and a who-does-what-where matrix across incidents.

Exercise prototype with fictional data. SMS and email are prepared and recorded; no live delivery gateway is connected.

Process flow diagram

One night in Kandy, start to finish

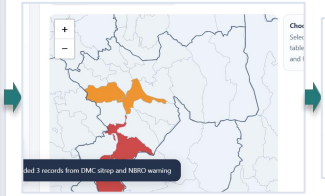
A fictional exercise, recorded in the real app with real Gemini.



1 Incident opened

District Secretariat

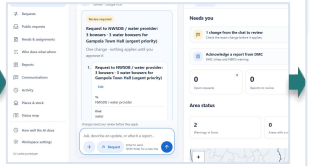
Every invited body is told at once.



2 Map turns red

DMC → District

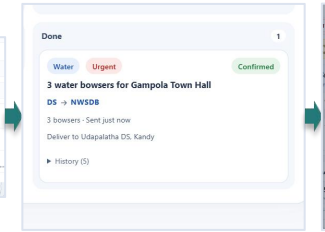
A pasted report turns divisions red and amber.



3 One sentence

Divisional Secretariat

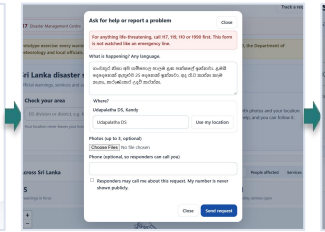
Gemini prepares the request to NWSDB.



4 Delivered

NWSDB → DS

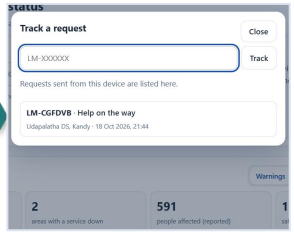
Accepted, delivered, confirmed.



5 Nimali asks

Resident, in Sinhala

25 people, no food tonight.



6 Help on the way

NDRSC → Resident

Her tracker shows each step.

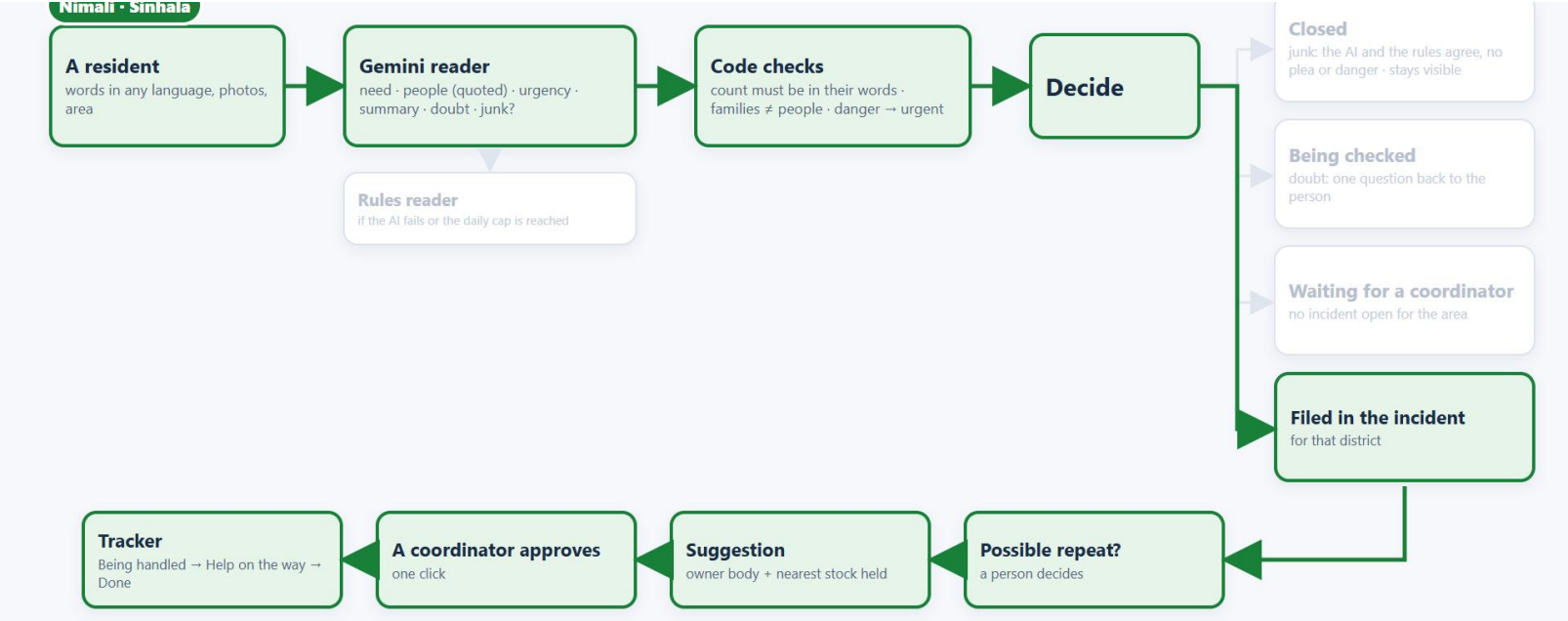
AI in this story: reads the report, prepares the request from one sentence, reads Nimali's Sinhala, suggests the request to the relief agency, and translates the warning texts. People approve each change.

Screens from the recorded exercise (local copy of the deployed build, real Gemini). Five teams in five browser tabs; roles are simulated.

Process flow diagram

How a public request is handled

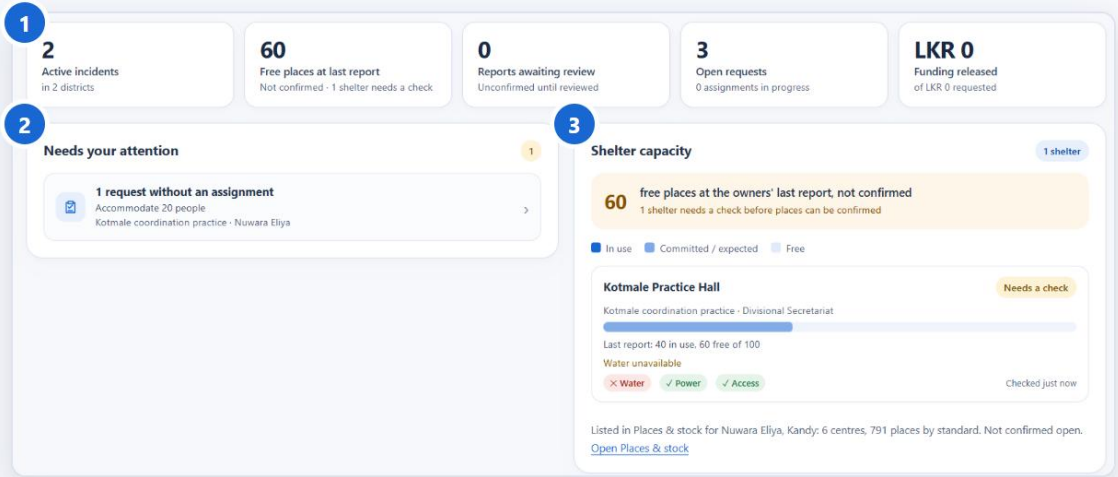
Green: Nimali's request, all the way to “Help on the way”. An advert would end in Closed, still visible.



Closing needs the AI and the rules to agree, with no plea or danger word. If the AI fails or the daily cap is reached, the rules reader takes over and says so.

Wireframes / mock diagrams

Each organisation's home: the Dashboard

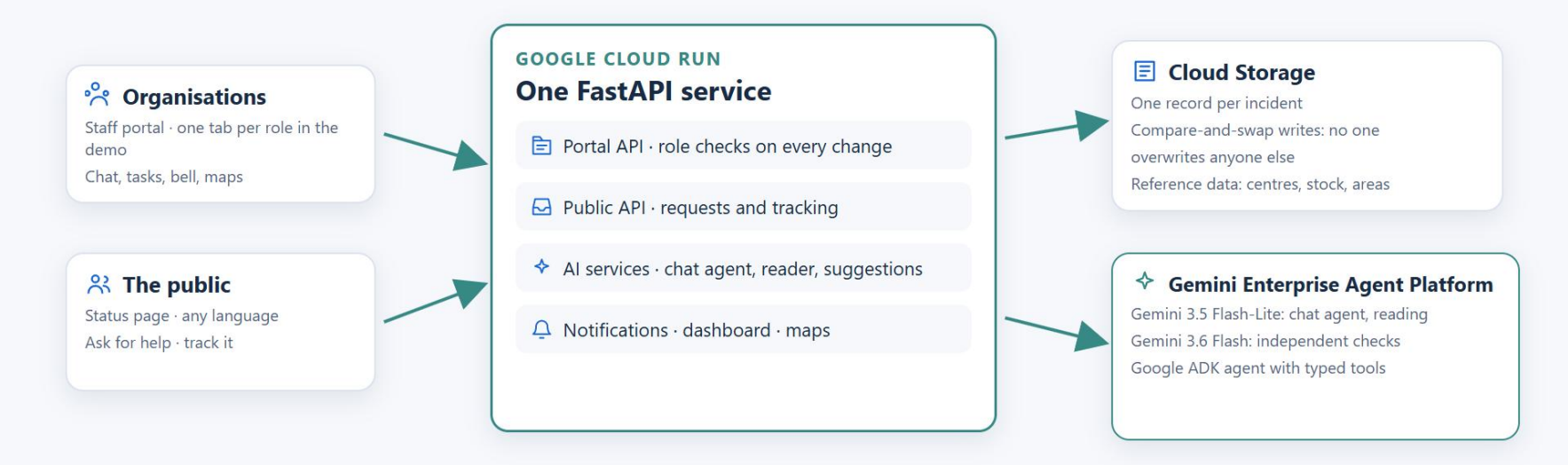


1. **Key figures**
   incidents, free shelter places, reports to review, open requests, funding
2. **Needs your attention**
   the next step for this organisation and role
3. **Shelter capacity**
   last reported places; unconfirmed numbers are marked as such

Real screen with numbered callouts. Below it: the live status map and every incident the organisation takes part in.

Architecture diagram of the proposed solution

Simple to run, safe to share

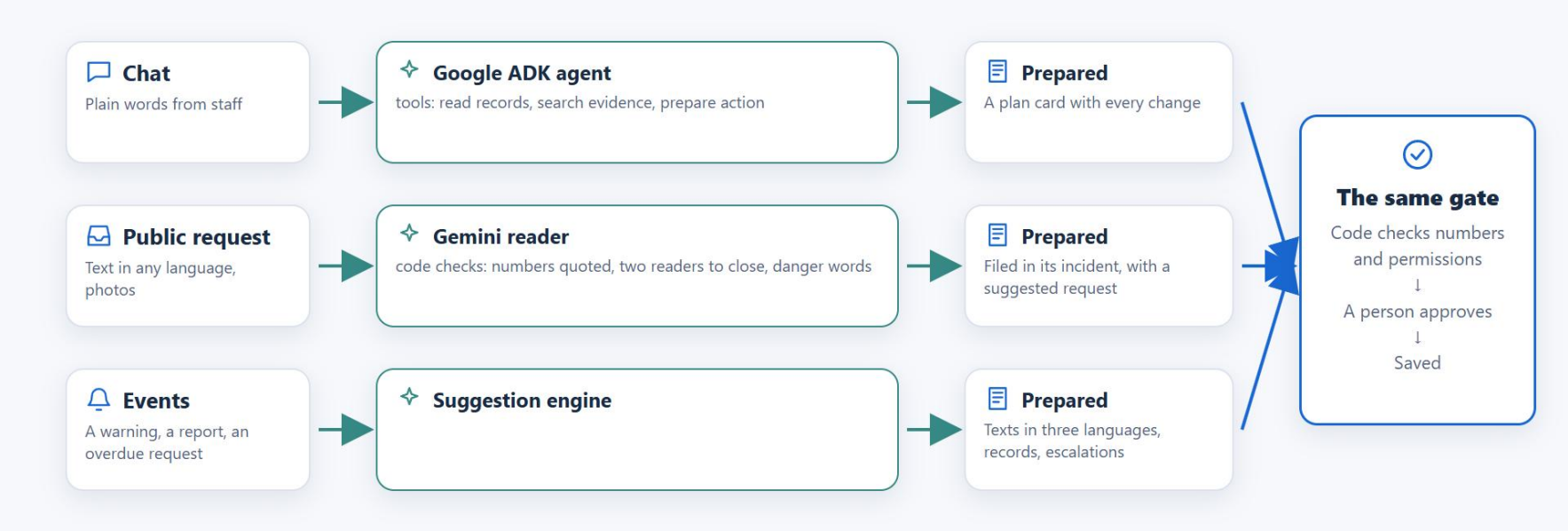


Every write: role and organisation checked on the server, the record version compared (compare-and-swap), the exact approved content re-checked. Demo roles are simulated; real accounts use sign-in.

Region asia-south1. Firebase and Firestore are not used. Cloud Storage is object persistence with generation preconditions, one writer today.

AI architecture

AI in three paths, one gate



Chat agent: Gemini 3.5 Flash-Lite in Google ADK. Reader: Gemini 3.5 Flash-Lite. Translations and independent checks: Gemini 3.6 Flash.

AI architecture · the chat agent

One chat turn, step by step

Blue: the road request as it really ran: read the incident, find places twice, prepare the request.



Each tool result is reviewed before the next step. This turn: 37,771 tokens in (70% cached), 201 out.

AI architecture · guardrails

Built to be trusted

A person approves every change
The AI prepares; nothing applies until approved

Tied to the exact version
Approval is bound to the plan's hash and the record version

The server re-checks
Role, organisation and permissions on every write

Numbers must be quoted
Every number must appear in the person's words or the records

Two readers to close
Gemini and the rules must agree; danger words are never closed

A bounded agent
Allow-listed tools; at most 12 model calls and 75 s a turn; plans of up to 8 steps

Text is data, not orders
Reports and public messages are never instructions to the AI

Fallback and audit
Rules take over if the AI fails, and say so; every turn is recorded

AI prepares. People decide. Every step is recorded.

AI architecture · models and measurements

The right model for each job, measured

AI task	Model	Measured per call	Cost per call	Guardrail
Chat agent (Google ADK, 10 tools)	Gemini 3.5 Flash-Lite	37,771 in (70% cached) · 201 out	≈ $0.012	≤ 12 model calls · 75 s · plans ≤ 8 steps
Public request reader	Gemini 3.5 Flash-Lite	668 in · 142 out	≈ $0.0006	Schema output; numbers quoted; rules fallback
Warning texts, Sinhala and Tamil	Gemini 3.6 Flash	67 in · 58 out	≈ $0.0005	Level, places and hotline locked
Report and bulletin reading	Code readers + chat summary	—	—	A person ticks every record
Suggestions on events	Code + the models above	—	—	Prepared for the right role; approved

Context: the agent sees the organisation, role, current incident version and only records that role may read. Reports and public messages are passed as data, never as instructions. Repeated context is cached, which cut this turn's input cost by 70%.

Measured 18 Oct 2026. List prices per 1M tokens: Flash-Lite $0.30 in / $2.50 out; 3.6 Flash $1.50 / $7.50. Cached input priced at full rate.

Technologies to be used in the solution

Google AI and Google Cloud, end to end

Technology	What it does for LastMile
Gemini 3.5 Flash-Lite · Gemini 3.6 Flash	Read requests and reports, plan actions, translate warning texts
Gemini Enterprise Agent Platform	Hosts the Gemini models (formerly Vertex AI)
Google ADK	The chat agent: a bounded loop over ten typed, allow-listed tools
Cloud Run (asia-south1) + FastAPI	Web app and API; role, scope and version checks on every write
Cloud Storage	One record per incident with compare-and-swap; reference data and photos
Cloud Build · Artifact Registry · Cloud Monitoring	Container builds, images, token and request usage for cost tracking
Leaflet · geoBoundaries (OCHA) · Lit · Web Awesome	Maps of 25 districts and their DS divisions; accessible interface components
pytest · Playwright · axe	621 automated API and browser tests; accessibility checks on 55 screens

Not used: Firebase, Firestore. Local development stores records in SQLite.

Estimated implementation cost

About a cent per AI action

The prototype today, measured, beside a real national deployment, estimated for a severe monsoon month.

≈ $3

per month today (measured prototype)

≈ $460

per month at national scale, severe month (est.)

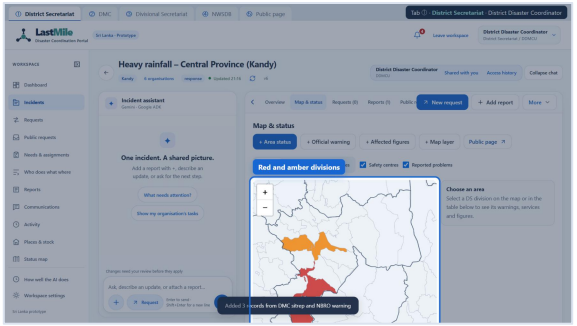
≈ $0.0006 per public request · ≈ $0.012 per chat turn

Resource	Prototype today (measured)	Real deployment (estimate)
Cloud Run	1 vCPU · 512 MiB · max 1 instance, scales to zero · ~4.5k requests/30 days → ≈ $0 (free tier)	2 always-on instances (1 vCPU · 1 GiB), more at peaks → ≈ $192
Gemini	Whole build so far: 1.07M Flash-Lite + 0.67M 3.6 Flash input tokens → ≈ $2.69	50k public requests ($28) + 20k chat turns ($237) + 4k translations ($2) → ≈ $267
Cloud Storage	4.7 MB of records → < $0.01	≈ 20 GB with photos → < $1
Images, builds, logs	27 images (shared layers), builds and logs in free allowances → < $1	Same → < $1
Total per month	**≈ $3**	**≈ $460 (quiet month ≈ $220)**

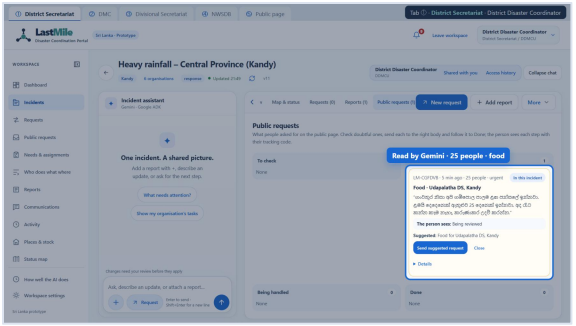
Upper bound: Cloud Run at higher-tier rates. Excludes SMS gateway charges, a transactional database (roadmap), staff and taxes.

Snapshots of the prototype

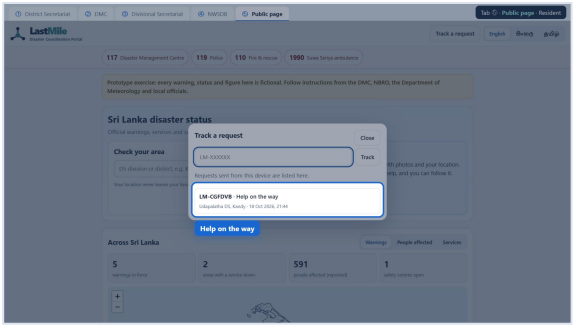
Real screens from the Kandy exercise



One live picture: red and amber divisions



Public requests board, read by Gemini



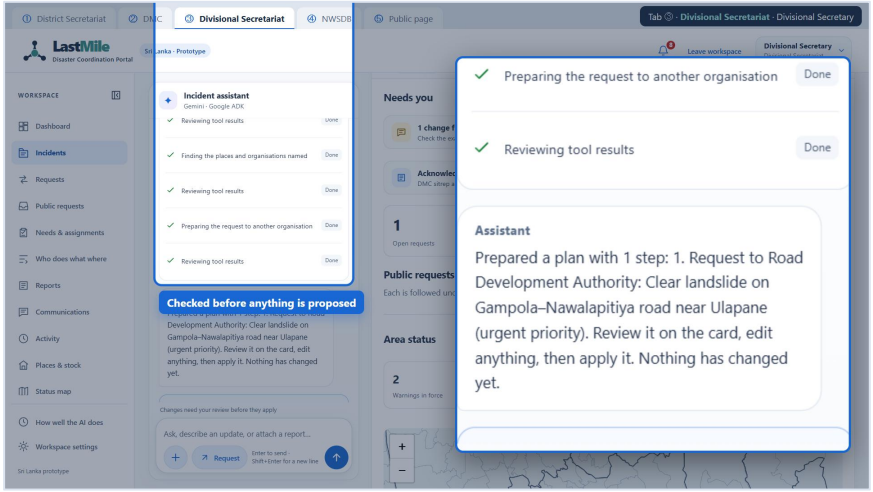
The resident's tracker: Help on the way

What the judges can try: open the live app, pick an organisation and role, ask for help on the public page in any language, and watch the request arrive on the board, be sent to the right body, and reach “Help on the way”.

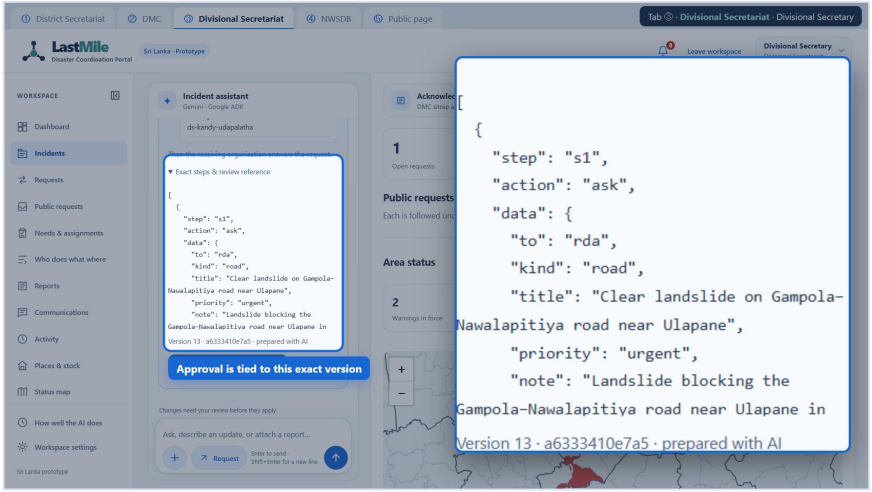
Recorded 18 Oct 2026 with real Gemini; fictional data.

Snapshots of the prototype

Inside the AI: every step is visible



The agent's steps: tool calls and the review of each result, shown in the chat.



The exact version: the plan's steps and its version and hash, approved as one.

Magnified from the recorded road request: Request to Road Development Authority, urgent, Udapalatha.

Prototype performance report / benchmarking

Measured, not assumed

92%

right decisions by Gemini

0

genuine requests wrongly closed

621

automated tests passing

Held-out set · 24 public requests	Rules alone	Gemini
Right decision (file, hold, check, close)	19/24 (79%)	22/24 (92%)
Right need	15/19 (79%)	19/19 (100%)
Right owning body	16/19 (84%)	19/19 (100%)
Urgent cases flagged urgent	4/4	4/4
Junk closed	0/3	2/3
Real requests wrongly closed (target 0)	0	0
Development set · 54 cases · right decision	51/54 (94%)	53/54 (98%)

Synthetic cases written by the team (English, Sinhala, Tamil, Singlish), shown live in the app under “How well the AI does”. Accessibility: 0 axe violations on 55 screens.

Additional details / future development

Next: from exercise to the field

01 Exercise with real responders

Timed exercises with DMC, district and divisional staff; native Sinhala and Tamil review; real report photos.

02 Close the loop with residents

Residents confirm or reopen “Done”; measure time from request to help.

03 Many writers, safely

A transactional database so many agencies can write at once, with the same checks.

04 Connect live channels

SMS gateway, verified agency feeds and outbound alerts, once agreements are in place.

Roadmap only. No agency partnership, field adoption, live alert delivery or payment is claimed.

Links

See it, try it, read it

Final product link

https://lastmile-alert-74y2iwqhzq-el.a.run.app

Choose an organisation and role; the public page is at /public.

Demo video link (3 minutes)

https://www.youtube.com/watch?v=t2vgS3u6gh8

LastMile Response, Team MOB · 3:23

GitHub public repository

https://github.com/LithiraAponsu2/lastmile-response-submission

Source, judge guide, architecture and evidence.

Prototype with fictional exercise data.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you



LastMile Response

Team MOB

Lithira Aponsu
Pulsara Aponsu

From the first warning,
to the last mile.